

Hanzhen Yu

hanzhen625@berkeley.edu | <http://www.linkedin.com/in/hanzhen-yu>

Education

University of California, Berkeley

Berkeley, CA

GPA: 4.0, Intended Data Science + Geography Major, GIST Minor

May 2029

- **Relevant Coursework:** Data Structures, Computational Structures, Data Science Foundations, Computer Simulations, Intro to GIS, GIS for Health Equity, Remote Sensing, Ecological Statistics

Projects

ArcGIS StoryMap - California Farm Workers Health Hazards

Nov. 2025 – Dec. 2025

Course Project - GIS and Spatial Analysis for Health Equity

Berkeley, CA

- Developed an ArcGIS StoryMap on California farmworker health risks, combining CalEnviroScreen datasets with PDF reports.
- Conducted GIS proximity and buffer analyses to map hospital access gaps and pesticide exposure risks near schools in farmworker communities.

Proximity Heat Map using QGIS

Nov. 2025 – Dec. 2025

Course Project - Intro to Geospatial Technologies

Berkeley, CA

- Designed proximity heat map of Gowen Cypress native habitat to human disturbances using QGIS.
- Recreated Gowen Cypress distribution map by merging a 1972 map with modern citizen-science databases.

Work Experience

University of Utah Summer Program for Undergraduate Research

May. 2026 – Aug. 2026

Undergraduate Researcher

Salt Lake City, UT

- Conducted GIS-based research on automated drainage basin delineation for alpine rock glacier.
- Built GrassGIS, ArcGIS Pro, and Python workflows to test DEM resolution and rooting-zone buffer effects across 178 basins, using area and overlap metrics to compare automated results with manual outlines.
- Currently contributing to a research manuscript based on the project findings.

UC Berkeley Studio.geo

Jan. 2026 – Present

Undergraduate Researcher

Berkeley, CA

- Clean and reconstruct GIS map layers of Bay Area creeks to visualize pre-European settlement hydrology for "Before You Are Here: Rematriating the Map in the Ohlone Bay Area", in collaboration with the Sogorea Te Land Trust.
- Conduct historical and topographic research to identify original flow channels, reconnect fragmented or diverted waterways, and restore accurate representations of creek systems altered by urbanization and drainage infrastructure.

Golden Gate Science Olympiad (GGSO)

Jul. 2025 – Jul. 2026

Executive Board Member, Finance/Outreach Director

Berkeley, CA

- Managed budgeting, income, and expenses in Google Sheets for a nonprofit organization that runs a high school Science Olympiad competition serving 1,000+ students.
- Secured \$5K+ in sponsorships from corporations and university departments through grant proposals and targeted email outreach.
- Founded GGSO Bay Area Urban Schools Initiative to offer free participation spots for 120 students new to Science Olympiad.

SKILLS & INTERESTS

Programming Languages: Python, Java, SQL, R

Software: ArcGIS Pro, QGIS, GrassGIS, Google Earth Engine, Microsoft Workspace, Adobe Products, Davinci Resolve

Languages: Proficient in Chinese (Mandarin)

Interests: Identifying plants/wildlife using iNaturalist, Top 100 global GeoGuessr player, Reading Brandon Sanderson's *Cosmere*, Outdoor enthusiast in hiking, skiing, and amateur mountaineering